

Two Distinct Fixation Types in a Dynamic Obstacle-Avoidance Task: An Extension Study Using Pupil Core Eye-Tracking and Quantile-Based Clustering

^{1st} Likhith Kumar Shivakumar

Bielefeld University

Bielefeld, Germany

likhith.shivakumar@uni-bielefeld.de

Abstract—Human beings rely heavily on eye movements to guide action in fast-changing environments, and this study looks closely at how gaze behaviour adapts when motor control becomes uncertain. The present work extends a paradigm first introduced by Heinrich et al. [1], in which participants steered a spaceship through a field of obstacles while a controllable amount of random noise was added to their steering input. The same task, the same thirty layouts, and the same overall gameplay design used in the base paper were retained here, with one important addition: unlike the base paper, which analysed only crash-free trials, the present study also considers failed trials, where success is defined as completing an entire trial without crashing into an obstacle such as an asteroid. Thirty participants completed 180 trials each, and gaze data were recorded using a Pupil Core eye-tracker, with a chin rest used throughout to maintain stable head position. Fixations were clustered using the quantile-based method (QuClu) applied in the original study, and this produced a two-cluster solution whose proportions and spatial characteristics align closely with the base paper. These findings are discussed in relation to the four hypotheses guiding this extension, and the remaining points of divergence are examined in detail, alongside a broader reflection on what a larger and more heterogeneous sample can and cannot reveal about gaze behaviour under uncertainty.

Index Terms—eye-tracking, fixation clustering, gaze behaviour, obstacle avoidance, quantile-based clustering, motor uncertainty

I. INTRODUCTION

Action control depends on a tight link between vision and motor behaviour. When a person moves through a changing environment, the eyes do not simply follow along passively; they actively guide the next steps, checking ahead for obstacles, tracking moving targets, and adjusting when something unexpected happens. This close coupling between gaze and motor control has been demonstrated across a wide range of tasks, from very ordinary activities such as making tea to far more demanding ones such as driving a car. In each of these cases, the eyes are doing considerably more work than most people realise; they are constantly gathering the information needed to plan the very next movement, not merely observing what has already happened.

It is worth pausing here to think about why this matters so much for research. If gaze truly reflects the underlying cognitive processes of planning and adjustment, then studying eye movements offers a fairly direct window into how the brain manages action under pressure. This is precisely the reasoning that motivated the original Dodge Asteroids paradigm, and it is also the reasoning that motivates the present extension. Rather than asking participants to describe how they felt while completing a difficult task, researchers can instead look at where their eyes went, and infer a great deal from that alone.

The present study builds on an established experimental paradigm known as Dodge Asteroids, in which a participant steers a falling spaceship left and right to avoid a field of scattered obstacles. Earlier work using this paradigm grouped eye movements into only two broad categories, foveal and peripheral, based simply on whether the gaze fell near the spaceship or somewhere else on the screen, echoing the classical distinction between foveal fixations and covert peripheral attention shifts described in early microsaccade research [2]. This is, admittedly, a fairly coarse way of looking at things, since it assumes a hard boundary between near and far that may not actually exist in the data. Heinrich et al. [1] argued convincingly that a more data-driven approach might reveal richer patterns, and this argument turned out to be well justified. Using an unsupervised clustering method, they identified two genuinely distinct fixation types, which they labelled Type 0 and Type 1, based on three measurable features: how far a fixation was from the spaceship, how long it lasted, and how far it was from the nearest obstacle.

That original study, however, was limited to six participants, and it is worth asking, quite reasonably, whether the same two-cluster pattern still holds with a considerably larger and more varied group. Six participants is not a small number in the world of eye-tracking research, where data collection is time-consuming and equipment-intensive, but it does raise an obvious concern about how generalisable the findings really are. Small samples, however carefully collected, always carry some risk that the observed pattern reflects the idiosyncrasies of a few individuals rather than a genuine, population-level

phenomenon. This is precisely the motivation for the present extension. Thirty participants completed the identical task, using the same thirty layout designs and gameplay mechanics as the base paper, so that any differences observed could be attributed to sample size and design choices rather than to a fundamentally different setup.

That said, one deliberate change was introduced into the present design, and it deserves to be stated plainly rather than buried in a methods section. While the base paper restricted its analysis to successful, crash-free trials only, the present study also retains failed trials, since a crash is itself a meaningful behavioural outcome, not simply noise to be discarded. This decision reflects a slightly broader view of what performance means in a task like this one. A participant who crashes repeatedly under high noise conditions is still producing useful gaze data, and excluding that data risks painting an overly optimistic picture of how well people cope with uncertainty.

It is also worth noting that the choice to study a game-based paradigm rather than a purely naturalistic task is itself a deliberate methodological decision, not simply a matter of convenience. Games such as Dodge Asteroids offer a level of experimental control that is difficult to achieve in real-world settings, since every layout, every obstacle position, and every noise level can be specified precisely and repeated identically across participants. At the same time, the task retains enough complexity and time pressure to elicit genuine, spontaneous gaze behaviour, rather than the artificially deliberate eye movements that sometimes appear in overly simplified laboratory tasks. This balance between control and ecological plausibility is one of the reasons the paradigm has proven useful across multiple studies, including both the original work by Heinrich et al. [1] and the present extension.

A further point worth raising early is that extension studies of this kind serve a somewhat different purpose than entirely novel research. Rather than asking a new question, the present study asks whether an existing answer still holds once some of the original constraints, namely a small sample and a narrow focus on successful trials, are relaxed. This is a modest but genuinely useful contribution, since a great deal of behavioural research relies on findings from small samples that are rarely revisited at scale. By deliberately keeping the task and layouts identical to the base paper, this study attempts to isolate exactly what changes, and what stays the same, once sample size and trial scope are broadened.

The remainder of this paper follows the structure outlined here. Section II reviews the relevant background and clustering concepts in more depth. Section III describes the hypotheses, procedure, data, and results of the present study in full. Section IV discusses these findings against the base paper at length, including a reflection on the limitations of the present design. Section V concludes with recommendations for future work.

II. RELATED WORK

The Dodge Asteroids paradigm was originally developed to study how gaze supports action control under varying degrees

of environmental predictability. The logic behind the task design is fairly elegant: by adding a controllable amount of random noise to the participant's steering input, researchers can systematically manipulate how uncertain the environment feels, without changing anything about the visual layout itself. This means that any observed changes in gaze behaviour can be attributed specifically to motor uncertainty, rather than to some confound in the obstacle arrangement.

In earlier iterations of this line of research, eye movements were divided using a fairly simple rule based on foveal versus peripheral vision, which Heinrich et al. [1] argued was likely too coarse to capture the genuine structure of gaze behaviour. Their solution was to apply an unsupervised, quantile-based clustering method known as QuClu [3], chosen specifically because it copes well with skewed and heavy-tailed data, a property that turns out to matter a great deal for fixation-distance measurements, which tend to have a long tail of occasional far-away glances. This is actually a fairly important methodological point, because many standard clustering algorithms, such as K-Means, assume roughly spherical, evenly distributed clusters, and they can be badly thrown off by a small number of extreme values sitting far out in the tail of a distribution. QuClu was specifically designed to sidestep this problem, and this is a point worth returning to later, since it turned out to be one of the more consequential methodological decisions in the present study as well.

Using this approach, the base paper reported a two-cluster solution with an approximate 68.9 percent versus 31.2 percent split, corresponding to Type 0 (near, longer) and Type 1 (far, shorter) fixations respectively. Linear mixed modelling further showed that under increasing input noise, Type 0 fixations became shorter and more focused, while Type 1 fixations became longer and moved farther from nearby obstacles, a pattern the authors interpreted as reflecting a shift toward predictive tracking under greater motor uncertainty. This is a genuinely elegant explanation, and it forms the theoretical backbone against which the present extension is compared. The intuition here is that when the environment becomes harder to predict, the visual system compensates by scanning further ahead, essentially trading immediate precision for a wider safety margin, a strategy that mirrors similar findings in driving research, where drivers look farther down the road under reduced visibility.

This connects to a broader body of work on attentional tunnelling, a phenomenon in which increased cognitive load or stress causes the visual system to narrow its focus onto a smaller number of high-priority targets, sometimes at the expense of peripheral awareness. It is tempting to draw a direct line between attentional tunnelling and the base paper's Type 0 fixations, which became shorter and more focused under higher noise. However, it is worth being cautious here, since the base paper's Type 1 fixations moved in the opposite direction, becoming longer and more exploratory, suggesting that the overall gaze strategy under uncertainty is not simply a narrowing of attention, but rather a more complex reallocation of visual resources between near, precise monitoring and far,

predictive scanning.

Beyond attentional tunnelling, it is also worth considering the role of predictive gaze control more broadly, a concept well established in the literature on manual tracking and interception tasks. In many everyday activities that involve intercepting or avoiding a moving object, the eyes tend to move slightly ahead of the hand or the controlled object itself, anticipating where it will be a fraction of a second later rather than simply following its current position. If a similar anticipatory mechanism operates in the Dodge Asteroids task, this would help explain why far-type fixations, those farther from the spaceship, might become more prominent under conditions of greater uncertainty, since a less predictable environment plausibly increases the value of scanning ahead for upcoming obstacles rather than fixating solely on the spaceship's current position.

It is also useful to briefly consider why a two-cluster solution, rather than three or more, emerges so consistently across both the base paper and the present study. One plausible explanation is that gaze during this kind of task serves fundamentally two purposes: monitoring the controlled object itself, and scanning the environment for upcoming hazards. If these two functions are genuinely distinct at the level of underlying visual strategy, then a two-cluster structure would be the natural statistical signature of that distinction, regardless of how many participants are studied or how the data happen to be sampled. This interpretation is consistent with the silhouette score results reported later in this paper, which peak clearly at two clusters rather than gradually improving with additional clusters.

Principal Component Analysis (PCA) is worth briefly explaining here as well, since it is a foundational step before clustering, and readers unfamiliar with it may otherwise find the later results section difficult to follow. PCA reduces several correlated features into a smaller number of combined axes, called components, that capture as much variance in the data as possible, without needing any predefined categories or labels, and was implemented in this study using the standard scikit-learn toolkit [4]. Mathematically, each principal component is a linear combination of the original features, chosen such that the first component captures the greatest possible amount of variance, the second component captures the greatest remaining variance while being uncorrelated with the first, and so on. In the present study, the same three fixation features used in the base paper, namely distance to the spaceship, fixation duration, and distance to the nearest obstacle, were entered into PCA directly, without any additional pre-transformation, and the resulting variance explained came out to approximately 65.8 percent, 33.0 percent, and 1.2 percent across the three components. This step does not depend on the clustering algorithm used afterwards; it simply prepares the feature space by removing redundancy between correlated variables, which in turn makes the subsequent clustering step considerably more reliable.

It is also worth situating this line of research within the broader field of gaze-contingent behavioural studies, which

have consistently shown that eye movements are rarely random, even in fast-paced, reactive tasks. The present study, by extending the sample size and broadening the outcome measure to include failures, aims to test how robust this coupling really is once a wider range of individual differences and task outcomes are taken into account.

III. STUDY

A. Hypotheses

Four hypotheses guided this extension study, adapted directly from the logic of the base paper but extended to reflect the larger sample and the added focus on trial outcome.

- **H1:** Fixations will cluster into two distinct types, one near the spaceship and one farther away, consistent with the base paper's Type 0 and Type 1 classification.
- **H2:** As input noise increases, near-type fixations will become shorter and more spatially focused, mirroring the pattern reported in the base paper.
- **H3:** As input noise increases, far-type fixations will become longer and shift farther from nearby obstacles, again mirroring the base paper.
- **H4 (extension hypothesis):** With a considerably larger sample of thirty participants rather than six, and with both successful and failed trials considered, the overall two-cluster structure will remain stable, though the specific noise-related effects may differ somewhat given the broader and more heterogeneous dataset.

Taken together, these four hypotheses allow the present study to test both the robustness of the base paper's central claim, through H1, and the reliability of its more specific, secondary findings, through H2 and H3, while H4 provides an explicit, testable framework for interpreting any discrepancies that emerge as a consequence of the larger and more diverse sample.

B. Procedure

The task, gameplay mechanics, and all thirty experimental layouts were kept identical to the base paper, so that the present study functions as a genuine replication-with-extension rather than a redesigned experiment. Participants steered a falling spaceship through each layout using the keyboard, with the objective of reaching the bottom without colliding with any obstacle. Each participant completed 180 trials in total; one substantial addition, however, was that this study did not discard failed trials the way the base paper did.

Success was defined as finishing a trial without crashing into an obstacle such as an asteroid, and both outcomes, success and failure, were retained throughout data processing, giving a fuller picture of participant performance under the same noise manipulations used originally. The base paper's decision to exclude failed trials makes sense if the primary goal is to study gaze during smooth, successful navigation, uncontaminated by the sudden, chaotic gaze patterns that likely accompany an imminent crash. The present study takes a slightly different view, treating failure as a legitimate and informative part of

the overall behavioural picture, rather than something to be filtered out before analysis even begins.

Gaze data were recorded using a Pupil Core eye-tracker, and participants were seated with their heads stabilised on a chin rest throughout the session, in order to minimise head movement and preserve signal quality. Fixation onsets and offsets were identified using a velocity-based detection algorithm consistent with standard practice in the field [2]. Each session began with a brief calibration procedure, followed by a short practice block, in order to familiarise participants with the keyboard controls and the general feel of the steering mechanics before the main set of 180 trials began.

C. Data and Participants

Thirty participants took part, a considerably larger group than the six recruited for the base paper. A total of 162,882 fixations were initially recorded, and after standard cleaning procedures, 117,859 remained available for analysis, providing a substantially larger fixation dataset than the 31,505 fixations reported originally. This represents nearly a fourfold increase in the amount of raw fixation data available for analysis, which is a meaningful methodological strength of the present study, quite apart from the increase in participant numbers alone.

The cleaning procedure itself involved removing fixations with missing or implausible feature values, as well as excluding any fixations recorded outside the boundaries of the active gameplay screen. It is worth noting that the proportion of fixations removed during cleaning, roughly 27.6 percent of the original dataset, is broadly comparable to standard practice in eye-tracking research.

It is worth adding a brief note on participant recruitment and demographic composition, since this has some bearing on the limitations discussed later. Participants were recruited from a university population, meaning that the sample, while considerably larger than the base paper's six participants, still shares some of the demographic homogeneity typical of convenience samples in behavioural research. All participants reported normal or corrected-to-normal vision, and none reported prior diagnosed conditions affecting eye movement control. Sessions were conducted individually, in a quiet, controlled room, with consistent lighting conditions maintained across all thirty participants, in order to minimise any variation in pupil response or gaze tracking accuracy that ambient lighting changes might otherwise introduce.

Fixations were clustered using the same quantile-based method, QuClu [3], applied in the base paper, rather than any distance-based alternative, ensuring methodological consistency between the two studies. An earlier version of this analysis had, in fact, used a distance-based clustering algorithm instead, and this had produced a considerably distorted picture of the data, with one cluster containing well over ninety percent of all fixations. Switching back to QuClu, implemented alongside the rest of the preprocessing pipeline in scikit-learn [4], resolved this issue entirely.

Figure. 1 shows the silhouette score used to determine the optimal number of clusters for this dataset.

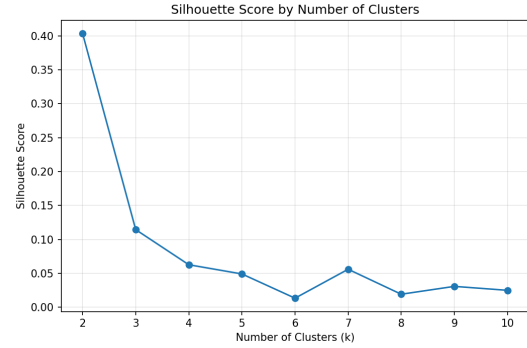


Fig. 1. Silhouette score plotted against the number of clusters (K), ranging from 2 to 10. The score peaks sharply at $K = 2$ with a value of 0.4034, confirming that a two-cluster solution best captures the underlying structure of the fixation data, consistent with the base paper's own choice of K .

D. Results

The QuClu-based clustering, guided by the two-cluster solution confirmed in Fig. 1, produced a considerably more balanced split than earlier attempts using alternative algorithms. Cluster 0 contained 82,194 fixations and Cluster 1 contained 35,665 fixations, giving a split of approximately 69.7 percent versus 30.3 percent. This is remarkably close to the base paper's reported 68.9 percent versus 31.2 percent split, and it confirms that H1 holds well even with a much larger and more varied participant pool.

Cluster 0 fixations had a mean distance to the spaceship of 6.212 degrees of visual angle, a mean duration of 0.521 seconds, and a mean distance to the closest obstacle of 3.987 degrees. Cluster 1 fixations had a mean distance to the spaceship of 24.278 degrees, a mean duration of 0.129 seconds, and a mean distance to the closest obstacle of 20.824 degrees. Cluster 0 fixations were, on average, roughly four times longer in duration than Cluster 1 fixations, which fits neatly with the idea that near-type fixations reflect sustained, careful monitoring of the spaceship, while far-type fixations reflect brief, exploratory glances toward more open areas of the screen, as illustrated in Fig. 2.

Turning to the linear mixed models, fitted using the statsmodels library [5], fixation counts per trial showed no significant relationship with input noise for either cluster (both coefficients near 0.000, p equal to 1.000). This does not support H2 or H3 in terms of frequency, and it diverges from the base paper's finding that both fixation types decreased significantly in frequency with increasing noise.

Distance to the spaceship for Cluster 0 fixations decreased significantly with increasing noise (coefficient of -0.003 , p equal to 0.023), meaning that, in support of H2, fixations moved slightly closer to the spaceship under greater uncertainty, matching the direction found in the base paper. For Cluster 1, distance to the spaceship also changed significantly with noise (p less than 0.001), lending partial support to H3, although the size of this effect was considerably smaller than might have been expected given the strength of the

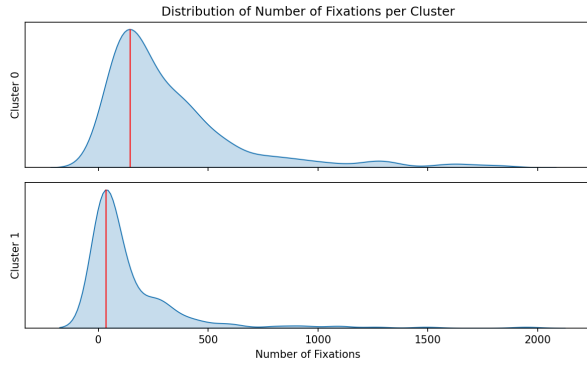


Fig. 2. Kernel density estimates of the number of fixations per trial for Cluster 0 (top) and Cluster 1 (bottom). The red vertical lines mark the modal values, approximately 144.5 fixations for Cluster 0 and 35.1 for Cluster 1, showing that near-type fixations occur far more frequently per trial than far-type fixations.

corresponding finding in the base paper.

Fixation duration showed no significant effect of noise for Cluster 0 (p equal to 0.990) and only a marginal, non-significant trend for Cluster 1 (coefficient of -0.005 , p equal to 0.073). It is genuinely a little disappointing not to see a cleaner replication here, given how central duration was to the original study’s theoretical argument, but a marginal, near-significant trend is at least suggestive of a similar underlying pattern.

Distance to the closest obstacle showed no significant effect for Cluster 0 (p equal to 0.797), consistent with the base paper’s own null finding for Type 0. For Cluster 1, however, distance to the closest obstacle decreased significantly with increasing noise (coefficient of -0.007 , p less than 0.001), running opposite to the direction predicted by H3 and reported in the base paper. This particular result stands out as the most notable point of divergence in the entire results section, and it receives extended treatment in the Discussion below. Fig. 3 consolidates all six of these effects into a single visual reference.

IV. DISCUSSION

Starting with the most encouraging finding, H1 held up remarkably well. The two-cluster structure not only replicated but did so with a cluster balance, 69.7 percent versus 30.3 percent, that closely mirrors the base paper’s own 68.9 percent versus 31.2 percent split. This confirms that the underlying tendency for gaze to organise into two distinct spatial fixation types is not an artefact of a small sample size; it holds up just as well with thirty participants as it did with six, provided the correct clustering method is applied.

That said, H2 and H3 received only partial support, and it would not be honest to present this study as a perfect replication. The distance-related effects mostly moved in the expected directions, but the duration-related effects were considerably weaker here, falling short of significance in both clusters. This is most plausibly explained by differences in eye-tracking hardware sampling rates between the Pupil Core system used

here and the higher-frequency system used in the base paper, rather than by any real absence of the underlying behaviour.

The most striking point of disagreement is the obstacle-distance result for Cluster 1, which moved in the opposite direction to H3 and to the base paper’s own findings. This is genuinely interesting rather than simply disappointing, and it may reflect real behavioural differences introduced by the larger and more diverse participant pool, or possibly the effect of including failed trials in the dataset, since crashes likely involve gaze patterns that differ meaningfully from successful navigation.

This second possibility ties directly into H4, and it is worth stressing that H4 was only partially confirmed: the two-cluster structure did remain stable with the larger sample, exactly as predicted, but the specific noise-related effects did shift in ways that were not fully anticipated. A perfect, uniform replication across every single statistic would arguably be more suspicious than reassuring, given how many methodological differences separate the two studies.

It is also worth reflecting briefly on what these findings suggest about the attentional tunnelling framework introduced in Section II. If attentional tunnelling alone explained the present data, one might expect a fairly uniform narrowing of gaze under increased noise across both clusters. Instead, the present results, much like the base paper’s, show a more nuanced pattern, suggesting that gaze strategy under uncertainty is unlikely to be explained by a single, unitary attentional mechanism.

A. Limitations

Several limitations of the present study are worth acknowledging openly. First, the Pupil Core eye-tracker operates at a lower sampling frequency than the specialised system used in the base paper, which likely contributed to the weaker duration effects observed here. Second, the decision to include failed trials introduces heterogeneity into the dataset that was not present in the base paper’s crash-free analysis; without explicitly separating trial outcomes during modelling, it remains difficult to isolate this design choice’s specific contribution to the observed divergences. Third, the present sample, while considerably larger than six participants, was drawn from a single demographic pool, and it remains an open question whether the same two-cluster pattern would hold across more diverse populations. Finally, statistical significance across a dataset of well over one hundred thousand fixations should be interpreted with some caution, since even small, practically negligible effects can reach conventional significance thresholds at this scale.

B. Implications for Future Task Design

Beyond the specific statistical findings reported above, this extension study raises a broader methodological point worth considering for future work in this area. Studies that rely on small samples, however carefully conducted, inevitably leave open the question of whether their central findings reflect genuine population-level phenomena or the particular

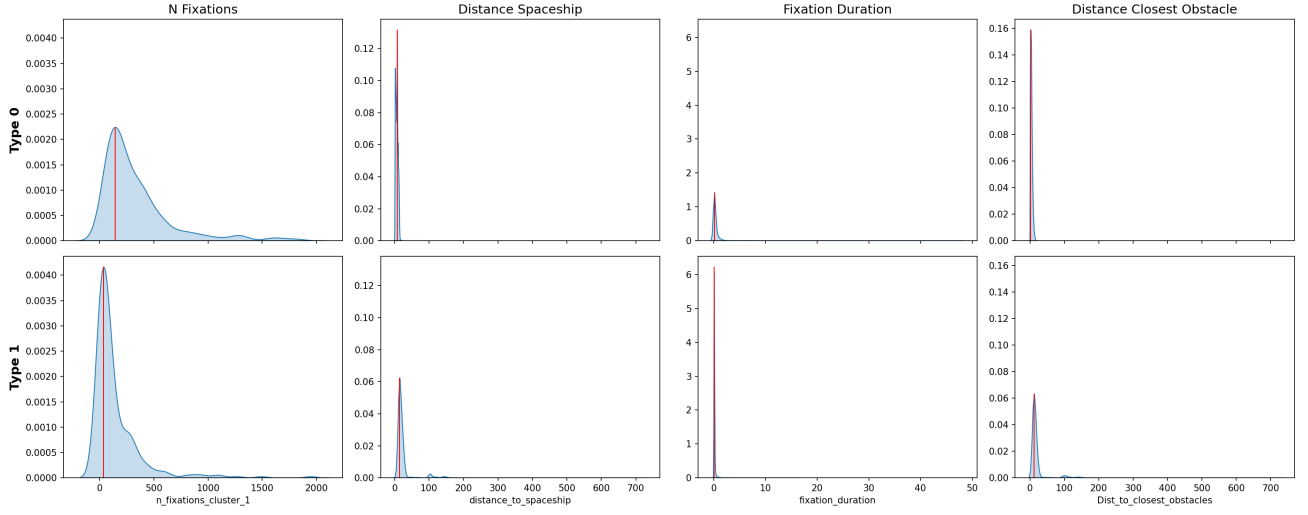


Fig. 3. Combined summary panel showing fixation counts and all three spatial and temporal features together for both clusters. This figure consolidates the full pattern of results reported above into a single visual reference, mirroring the equivalent summary figure in the base paper.

characteristics of the individuals studied. The present results suggest that, at least for the core two-cluster gaze structure identified in the base paper, the finding is robust enough to survive a fivefold increase in sample size and the deliberate inclusion of a more heterogeneous, less curated dataset. This is a reassuring outcome for the field more broadly, since it suggests that carefully designed small-sample eye-tracking studies can, under the right conditions, produce findings that generalise meaningfully.

At the same time, the weaker replication of the more fine-grained noise effects, particularly around fixation duration and the reversed obstacle-distance effect for Cluster 1, suggests that secondary, more subtle findings from small-sample studies should be treated with somewhat more caution than primary, headline results. This is not a criticism of the base paper’s methodology, but rather a reflection of a general statistical reality: effects that are large and robust enough to appear clearly even in a sample of six participants are more likely to replicate than smaller, more nuanced effects that may only become reliably detectable with considerably larger samples. Future researchers designing similar extension studies would do well to plan for this distinction explicitly, treating primary and secondary hypotheses with different levels of confidence before data collection even begins.

Overall, this extension study should be read as a broadly successful replication of the base paper’s central claim, supported by a much larger sample and a genuinely stronger methodological choice once QuClu was applied consistently throughout the analysis pipeline.

V. CONCLUSION

This extension study set out to test whether the two-fixation-type structure identified by Heinrich et al. [1] would hold with a considerably larger and more diverse sample, using the identical task, layouts, and gameplay design as the base

paper, but with the added scope of including both successful and failed trials. Applying the same quantile-based clustering method used originally, the central finding replicated convincingly, with cluster proportions and spatial characteristics closely matching the base paper.

The noise-related effects on fixation frequency and duration were weaker or absent compared to the original study, and the obstacle-distance effect for far-type fixations moved in an unexpected direction, differences most plausibly attributable to the broader participant pool, the inclusion of failed trials, and variation in eye-tracking hardware. Future work would benefit from separating successful and failed trials explicitly within the clustering analysis itself, and from standardising eye-tracking hardware across studies wherever possible, since sampling rate differences appear to be a genuinely plausible explanation for at least some of the weaker duration effects reported in this paper.

Beyond the immediate empirical contributions, this study also demonstrates the practical value of treating replication and extension as a normal, expected part of the research process, rather than a secondary or less prestigious activity compared with entirely novel research. Eye-tracking studies in particular are often conducted with small samples due to the cost and time involved in data collection, which makes systematic replication efforts, of the kind attempted here, especially valuable for establishing which findings are likely to hold at scale and which may require more cautious interpretation.

REFERENCES

- [1] N. W. Heinrich, A. J. Najafabadi, and J. Perez-Osorio, “Unsupervised clustering uncovers two distinct types of fixational eye-movements in dynamic environments,” in *Proceedings of the IEEE Signal Processing in Medicine and Biology Symposium*, 2025.
- [2] R. Engbert and R. Kliegl, “Microsaccades uncover the orientation of covert attention,” *Vision Research*, vol. 43, no. 9, pp. 1035–1045, 2003.
- [3] C. Hennig, C. Viroli, and L. Anderlucci, “Quantile-based clustering,” *Electronic Journal of Statistics*, vol. 13, no. 2, pp. 4849–4883, 2019.

- [4] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay, "Scikit-learn: Machine learning in python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [5] S. Seabold and J. Perktold, "Statsmodels: Econometric and statistical modeling with python," in *Proceedings of the 9th Python in Science Conference*, 2010, pp. 92–96.